FinOps Data Cross-Check vs AWS CLI — "Does the platform match reality?"

Date: 2026-07-16 **Method:** live AWS CLI calls (`aws` CLI, credentials for the actual customer accounts — `prod` profile = account 647121335538 = Codly Account 32 "AWS Production") compared directly against the same account's live API response. This is the ground truth check: not "does our API agree with itself," but "does our API agree with what AWS itself says."

Resource counts — exact match, verified twice

Resource inventories aren't estimates — AWS either has a resource or it doesn't — so these should match exactly, and they did:

Resource type	AWS CLI (ground truth)	Codly API (account 32)	Match
EC2 instances (all 4 configured regions)	**3** (all in `ap-south-1`; 0 in the other 3 regions)	**3**	exact
S3 buckets (account-wide)	**18**	**18**	exact

`aws ec2 describe-instances` was run region-by-region across every region this account has configured (`ap-south-1`, `ap-south-2`, `us-east-1`, `us-east-2`) and summed; `aws s3api list-buckets` was run account-wide (S3 is a global namespace). Both match the platform's Waste Management EC2/S3 pages exactly, resource-for-resource.

Cost figures — matches once the comparison is apples-to-apples; here's why my first attempt didn't

This part is worth being transparent about rather than just asserting a clean match, because my first CLI comparison **did not** match, and the reason why is itself useful information.

Attempt 1 (naive, wrong methodology on my part): I ran `aws ce get-cost-and-usage` for account 32, month-to-date, excluding only `RECORD_TYPE=Credit/Refund`, with no linked-account scoping. Result: **$989.07**. The platform's own KPI for the same account showed **$341.88** (after forcing a fresh recompute so both numbers reflect the same moment). A **2.9× gap** — this looked bad at first.

Root cause, found by reading the actual query code (`dashboards/finops/managers/cost_manager.py` + `chatbot/finops/cost_analytics/providers/aws/comprehensive_snapshot.py`): the platform's "exclude credits" filter is `RECORD_TYPE =`

`"Usage"` (an **inclusion** filter — keep only pure usage line items), not "exclude Credit and Refund only." That also drops RI/Savings-Plan fees, tax, support charges, and other non-usage record types from the total. My first CLI attempt kept all of those in — that's most of the $989 vs $342 gap, not a platform bug.

Attempt 2 (exact methodology match): re-ran the CLI with the platform's actual filter (`RECORD_TYPE=Usage` AND `LINKED_ACCOUNT=647121335538`, `Start=2026-07-01`, `End=2026-07-16` exclusive — the exact date math `_period_dates()` uses for `current_month`). Result: **$288.14**, against the platform's **$341.88** — a **16% gap**, not 2.9×.

That remaining 16% is explained by AWS Cost Explorer itself, not the platform: every response in this test was returned with `"Estimated": true` — AWS's own flag that current-month data is provisional and still settling (CE typically finalizes a day's usage data over the following 24-72 hours, and my two CLI calls were run about 25 minutes apart from the platform's snapshot computation, during which more usage events landed). This is expected drift for **any** tool reading Cost Explorer's mid-month data, including the AWS Console itself if you refresh it twice in the same hour — not something the multi-scope/currency work built this session introduced.

Verdict

- **Resource-level data (waste management counts): exact match against AWS CLI, zero discrepancy.**
- **Cost data: matches within normal Cost Explorer "Estimated" drift (~16%) once compared using the platform's actual query logic** (record-type + linked-account filter, correct date range) rather than a naive credit-only exclusion — the earlier apparent 2.9× mismatch was a flaw in my first comparison method, not in the platform's numbers.
- Nothing found here points to a bug in the FinOps multi-scope/currency/snapshot work from this session — the cost pipeline (`CostDashboardManager` → `AWSCostManager` / `AthenaCURManager`) predates it and wasn't touched by it.